

Bangzhe Huang

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Current position: Research Intern in Prof. Ziming Liu's group, College of AI, Tsinghua University

Research interests: physical world models, scientific foundation models, neural architecture search, AI4AI, and automated research

Education

Fudan University, Department of Physics

Undergraduate, Strong Foundation Program in Physics

Shanghai, China

September 2026 – expected June 2030

Research Experience

AI4AI: Learning to Design a Model

Lead researcher; advisor: Prof. Ziming Liu, College of AI, Tsinghua University; ongoing research

- Matched or exceeded published **SOTA scores on 33 forecasting tasks across 9 OpenTS datasets**, with up to **31.56% lower MSE** than the published best score.
- Designed a neural architecture search space rich in high-performing models, enabling straightforward random search and performance-guided mutation to discover strong architectures.
- Built a scalable search and evaluation pipeline across **32 A100 GPUs**, integrating architecture generation, automated training, feedback-guided search, and experiment tracking.

Scaling Foundation Models for Many-Body Physics

Project lead; advisor: Prof. Jing Wang, Department of Physics, Fudan University; preprint in preparation

- Built and trained a **Hamiltonian-conditioned foundation model** for condensed-matter physics spanning transverse-field Ising, Heisenberg, XXZ/XYZ, and Dzyaloshinskii–Moriya interactions.
- Led the end-to-end foundation-model engineering stack: physical-system representation, multi-task data generation, multi-GPU JAX training, mixed precision, resumable checkpoints, and real-time experiment monitoring.
- Integrated mixture-of-architectures feature fusion with neural quantum-state learning and physics-based evaluation to model ground-state wavefunctions across thousands of Hamiltonians.

Open-Ended Scientific Research Benchmark for Physics Research Agents

Independent research; academic discussion: Prof. Siheng Chen, School of Artificial Intelligence, Shanghai Jiao Tong University; pilot completed

- Designed open-ended physics research tasks with stable ground truth defined by low-dimensional physical structures and equivalent observables, without prescribing a single valid research trajectory.
- Built an end-to-end pilot spanning experiment and simulation design, code generation and execution, multi-round observation and feedback, and a final physical audit.

Phase Transport and Defect Dynamics in Nonreciprocal Spin Systems

Lead researcher; advisor: Prof. Zi Cai, School of Physics and Astronomy, Shanghai Jiao Tong University; pilot completed

- Used Monte Carlo simulations to study the microscopic origin of macroscopic oscillations in two-dimensional nonreciprocal spin systems and extracted oscillation frequency, domain-wall density, and defect observables.
- Connected oscillation frequency ω , domain-wall density ρ , and effective transport velocity through $\omega = \rho v_{\text{eff}}$, and extracted v_{eff} from the linear ω - ρ regime.
- Identified regimes dominated by vortex-pair generation and periodic wall expansion versus vortex-constrained topological structures, and tested the physical picture across lattice geometries and symmetric bond disorder.

Honors

Gold Medal, 42nd Chinese Physics Olympiad, 2025

Technical Skills

Machine learning: JAX, PyTorch, foundation-model training, neural architecture search, multi-GPU training, LLM agents

Scientific computing: Python, Monte Carlo simulation, numerical experiments, statistical analysis

Physics: quantum mechanics, statistical physics, many-body physics